

B02 Practical Mastery Check

Closed-book transfer - non-gating - save a genuine attempt first

Non-gating closed-book mastery check

This check does NOT unlock, block or replace Gate A. Save a genuine attempt before opening its protected review.

Scenario

Build a fresh monthly-sales batch using only the supplied Practice_Data folder. Do not open the learner book or protected review while attempting it.

- Write the input/output contract and the exact file-selection rule. Discover only sales_2026_*.csv in deterministic order and retain source_file lineage.
- Load the three monthly files, reconcile per-file and combined row counts, prove order_id uniqueness, and recompute/validate net_sales rather than trusting it blindly.
- Validate customers.csv before merge. State the intended cardinality and demonstrate what your pipeline does if customer_id is not unique. Do not silently dedupe.
- Declare the merge contract and unmatched-key check. Prove that a critical master-key validation failure blocks trusted output.
- Validate api_response.json as an API-shaped offline response: required top-level keys, metrics list shape, and expected field types. A live HTTP call is optional, not required.
- Create logging/validation evidence and a run manifest that records runtime versions, selected input files, row counts, key checks, critical check status and intended output path.
- Define the rerun policy. Show what happens on a second run and why it cannot silently duplicate data.
- Explain which environment/package/CLI/API-pagination/incremental concerns remain for Course 3 rather than pretending B02 covers them.

Evidence to save before review

- Your code/notebook or script attempt.
- Selected file list and per-file/combined reconciliation.
- Master-key uniqueness result and merge validation result.
- Machine-readable validation summary plus log/run manifest.
- Evidence that trusted output is withheld on critical failure.
- Second-run evidence.
- A 60-90 second own-words explanation of grain, failure containment and rerun behavior.

Assessment rule

Do not score this by percentage. Mastery requires the core engineering controls to be independently demonstrated with no unresolved critical failure. If you get stuck, save the failed/developing attempt before opening the protected review.